

Adjoint measurement and calibration of Manning roughness in E3SM’s river dynamical core

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Abstract

This paper presents a transient discrete adjoint of RDycore, the river dynamical core of the Energy Exascale Earth System Model (E3SM), and uses it to measure how much of a flood model’s error its roughness parameter is able to remove. We differentiate the production solver in place rather than rewrite it in a differentiable-programming framework: one exact, assembled Jacobian of the shallow-water right-hand side serves the implicit time integrator, the adjoint, and the calibration. An objective and its gradient cost about two forward solves regardless of the number of parameters, every derivative passes a finite-difference gate on the full nonlinear solve, and the whole gradient path runs on GPUs, where four NVIDIA A100s run a calibration iteration on a dense-gauge twin $19\times$ faster than a 64-core CPU node.

For Hurricane Harvey on a 2.93M-cell, 30 m mesh of the Houston area, scored against 46 surveyed high-water marks, the ceiling of Manning roughness is measurable before any optimizer runs. A single factor on the whole roughness field removes at most 0.08 m of a 0.72 m error, and only at roughness values below any entry in the lookup; fifteen land-cover classes calibrated separately remove 0.107 m, 15% of the error. The spectrum of the prior-preconditioned Gauss–Newton Hessian, from sixteen forward runs, shows why: at a $\pm 30\%$ width on the land-cover lookup, the marks determine about three combinations of the fifteen classes, and calibrating those three recovers 84% of what fifteen achieve. What the three reach is the same at every prior width we ran: developed-medium and woody wetland at $0.3\times$ their lookup values, below the lookup’s value for developed open land. The calibrated friction is compensating for an error that is not in the friction. What a $\pm 30\%$ prior cannot support is the fifteen-class field. The survey is the observable because at seven of twelve stream gauges the 30 m cell’s bed sits above the peak observed water surface: the mesh holds the bank of an incised bayou, not its channel. Calibrating the same three classes on the gauges that do hold water sends them the opposite way, to the top of the prior instead of the bottom, and the resulting field scores worse at the marks than no calibration at all. Roughness is absorbing each observable’s own error. Pointed at the initial state, the same measurement finds more: a 20% reduction in stored water is worth 0.075 m. Most of the remaining error lives in the model’s water balance, not in its friction.

1 Introduction

Manning roughness is the dominant tunable parameter of depth-averaged flood models, and its calibration is traditionally manual or ensemble based. Adjoint methods deliver the gradient of an observation misfit with respect to *every* cell’s roughness at a cost that does not grow with the

number of parameters — a few forward solves, measured in Section 2.3 — making distributed calibration tractable. The approach is well established in shallow-water modeling, from channel networks [1] and unstructured finite-volume *twin experiments* — calibrations against observations the model itself generated from a known roughness field, so that the recovered field can be scored against a truth, and unrelated to the “digital twin” of a physical asset [2] — to GPU-accelerated optimizers [3]. It is deployed in variational-assimilation platforms such as DassFlow [4, 5, 6] and the Thetis coastal model, which calibrated a spatially varying Manning field for the Baltic Sea with an automatically generated adjoint [7], following storm-surge sensitivity studies that addressed the overfitting inherent in distributed friction estimation [8]. The closest published work to ours is Pujol et al. [9], who infer distributed friction in a 2D urban flood model by variational assimilation of multi-source data *including high-water marks*, and use adjoint-derived global sensitivity measures to decide how to spatialize the friction field before assimilating — the same observable and the same design move that Sections 6.1 and 6.2 employ, at street scale and with a Tapenade-generated adjoint of a solver built for assimilation. What this paper adds is neither the adjoint nor the observable but the setting and the question: a production Earth-system component differentiated in place, and a measurement of how much of the model’s error the parameter can explain at all. For this event and this observable the answer is 15%, and the spectrum of the calibration problem explains it: the marks determine about three roughness combinations, so a calibration of three parameters recovers most of what fifteen achieve and a wide prior releases the twelve it cannot determine. The measurement is not specific to roughness, and Section 7 turns it on the model’s initial state, where it finds more.

A recent generation of *differentiable* flood and hydrology codes obtains gradients by rewriting simplified solvers inside machine-learning frameworks: Inunda (PyTorch, local-inertial, calibrated on a Hurricane Harvey hindcast) [10], Hydrograd (Julia) [11], AegirJAX [12], and differentiable routing [13]. These demonstrate demand for exactly this capability, but none is a production Earth-system component: they trade the validated numerics, unstructured meshes, and coupling infrastructure of codes like RDycore for differentiability. A parallel line ports production hydrodynamics to GPUs without differentiating them — CaMa-Flood-GPU [14] is a GPU reimplementations of CaMa-Flood, with gradient-based calibration named as future work rather than demonstrated.

This paper takes the complementary route: make the production code differentiable. That route has a long precedent outside E3SM — most prominently MITgcm, whose exact adjoint is generated by automatic differentiation of the production ocean model [15] and underpins two decades of ECCO state and parameter estimation [16]. Ours differs in how the derivative is obtained rather than in ambition: the Jacobian is hand-differentiated and assembled, which makes it available to the implicit solver as well as to the adjoint, and Section 4 argues that for a differentiable friction term the implicit solve is not optional. Inside E3SM the precedent is narrower: the land-ice component MALI [17] routinely inverts for basal friction with an adjoint obtained by automatic differentiation of its Albany velocity solver [18] — but that is the adjoint of a *steady* nonlinear solve, used for initialization; no E3SM component has, to our knowledge, a discrete adjoint of its transient time stepper. RDycore is a finite-volume shallow-water dynamical core (Roe fluxes, well balancing, wetting and drying) validated at up to 471M cells. We add the missing derivative infrastructure — an exact assembled Jacobian and TSAdjoint-based sensitivities [19] — entirely behind a configuration flag, with no impact on existing runs and no new dependencies.

This paper contributes:

- an exact Jacobian of the shallow-water right-hand side — Roe dissipation, entropy-fix and wet/dry branches, and reflecting, Dirichlet and critical-outflow boundaries — verified against finite differences to 10^{-6} (Section 2.2.1);

- adjoint gradients with respect to the initial state and the roughness field, at multiple observation times, verified to 10^{-5} or better (Section 2.2.2);
- implicit backward-Euler stepping and its adjoint from the same Jacobian, with the measurement that the explicit friction term returns a non-finite state at every step tested on the 30 m Hurricane Harvey mesh (Section 4);
- a measurement of when an observing network makes a roughness field identifiable at all, from twin experiments and from the spectrum of the prior-preconditioned Gauss–Newton Hessian, which counts the roughness combinations the observations determine (Sections 5 and 6.4);
- the ceiling of roughness on the Hurricane Harvey survey error, 15%, and the three-parameter calibration the spectrum selects (Sections 6.2–6.4);
- a cross-observable validation: the three classes calibrated on the stream gauges move to the opposite edge of the same prior and score worse at the high-water marks than the uncalibrated lookup, which makes the survey-only result a fit rather than a skill (Section 6.1);
- the same measurement applied to the initial state, where it finds more, and the diagnosis of the downstream reach as a ponded lake behind the domain’s outlet (Sections 6.1 and 7).

2 Methods

2.1 Forward model

RDycore solves the 2D shallow water equations in conservative variables $u = (h, q_x, q_y)$, $\mathbf{q} = h\mathbf{v}$:

$$\partial_t h + \nabla \cdot \mathbf{q} = Q_r(x, t), \quad (1)$$

$$\partial_t \mathbf{q} + \nabla \cdot \left(\frac{\mathbf{q} \otimes \mathbf{q}}{h} + \frac{gh^2}{2} \mathbb{I} \right) = -gh \nabla z_b - g n^2 h^{-7/3} \mathbf{q} \|\mathbf{q}\|, \quad (2)$$

with cell-centered finite volumes on an unstructured mesh:

$$\frac{du_i}{dt} = -\frac{1}{|\Omega_i|} \sum_{e \in \partial\Omega_i} \mathcal{F}(u_i, u_j, \mathbf{n}_e) L_e + S(u_i; n_i, t) \equiv f_i(u, n), \quad (3)$$

where $\mathcal{F}$ is the Roe flux with a critical-flow (entropy) fix, primitives are reconstructed with a regularized division and a dry cutoff $h_{\min}$, and S contains bed slope, Manning drag, and external forcing.

The Roe flux’s dissipation is built from the moduli $|\lambda_p|$ of its characteristic wave speeds, so the closed-form expression it evaluates switches wherever a wave speed changes sign. Switches like this, *branches* in the sense that $|x|$ has a positive and a negative branch, recur throughout the discretization. They belong to the model rather than to the derivative: a nonlinear solver must accommodate them whether or not anyone differentiates the code, which is the argument of Section 4. Away from a switching surface the map is smooth and equals its active branch; the derivative machinery of Sections 2.2.1–2.2.2 is *branch-faithful*, differentiating the expression the condition actually selected to give the correct one-sided derivative.

A differentiable right-hand side. RDycore’s existing friction treatments (a semi-implicit splitting and the implicit scheme of Xia and Liang) embed the time step, and in the semi-implicit case the flux divergence, inside the RHS callback. The resulting map is consistent only with forward Euler and admits no well-defined $\partial f/\partial u$. We therefore added a plain explicit source treatment (`source.method: explicit`), used on every adjoint path, and Section 4 shows how the implicit integrator, rather than the right-hand side f , then handles the resulting stiffness.

2.2 Differentiating the production solver

Two design decisions shape everything downstream. We obtain the derivative of the right-hand side by hand, as forward-mode differentiation of the flux computation itself, branch by branch, and we *assemble* it into a sparse matrix rather than apply it matrix-free. Assembly is what lets one artifact serve three consumers: the implicit solver preconditions with it (Section 4), `TSAdjoint` transposes it, and the calibration couples it to the parameter Jacobian. The two parts below build that Jacobian (Section 2.2.1) and the adjoint sensitivities on top of it (Section 2.2.2); every derivative is gated by finite differences of the full nonlinear solve.

2.2.1 Exact Jacobian

The RHS Jacobian is block sparse with 3×3 blocks on the cell-adjacency graph: each interior edge (i, j) contributes $\mp(L_e/|\Omega_{i,j}|) \partial \mathcal{F}/\partial u_{L,R}$ to the two cell rows, local sources contribute diagonal blocks, and boundary edges contribute diagonal blocks through the ghost-state map of the boundary condition.

Edge orientation is a classical source of sign errors in this construction, so one consequence of that structure needs stating. The pair $(\partial \mathcal{F}/\partial u_L, \partial \mathcal{F}/\partial u_R)$ is evaluated *once* per edge, in the edge’s own stored normal direction, and then scattered to both cell rows with opposite area weights $-L_e/|\Omega_i|$ and $+L_e/|\Omega_j|$: the downwind cell receives the same two blocks with the roles of u_L and u_R exchanged between diagonal and off-diagonal positions. The normal is never flipped and the flux routine is never re-entered with reversed arguments, so conservation is structural rather than something the signs must reproduce, mirroring how the right-hand side itself accumulates the flux. The assembled-matrix checks in Table 1 would expose any lapse regardless, since they compare the full assembly against a matrix-coloring finite-difference Jacobian of the actual RHS on unstructured meshes with mixed boundary conditions, where a single mis-oriented edge is visible.

Flux blocks by forward-mode differentiation. Rather than deriving the dissipation-matrix derivatives symbolically, we differentiate the flux *computation*: a hand-written forward-mode (tangent) version of the Roe flux routine propagates directionals through every intermediate (Roe averages, eigenvectors, eigenvalues including both branches of the critical-flow fix, wave strengths, and physical fluxes), taking the code’s own branch at every $|\cdot|$ and switch. The result is a consistent one-sided derivative. Seeding unit directions through the primitive-reconstruction chain rule yields the two conservative-space blocks in six tangent evaluations per edge. The tangent code mirrors the flux code line by line, so it can be reviewed against it, and we found it correct on first assembly against the verification harness below.

Boundary blocks. For a boundary edge, $F(u_{\text{in}}, u_g(u_{\text{in}}))$ contributes $(\partial_L F + \partial_R F \partial u_g/\partial u_{\text{in}})$. The reflecting map is a linear mirror; Dirichlet ghosts are constant; critical outflow has a closed-form ghost map $h_g = (q^2/g)^{1/3}$ with the code’s inflow branch producing an identically zero flux.

Check	Configuration	Result	Gate
Edge flux blocks	wet jump 10 m/5 m	7.6×10^{-9}	10^{-6}
Edge flux blocks	transcritical (entropy-fix branch)	2.4×10^{-10}	10^{-5}
Source block	shallow flowing state	$< 10^{-6}$	10^{-6}
Assembled matrix	uniform flow, reflecting BCs	1.6×10^{-8}	10^{-6}
Assembled matrix	dam break, reflecting BCs	1.5×10^{-8}	10^{-6}
Assembled matrix	dam break, Dirichlet + critical outflow	1.9×10^{-8}	10^{-6}
dJ/du_0 (explicit RK)	8 sampled components	1.0×10^{-8}	10^{-5}
dJ/dn (explicit RK)	domain aggregate	2.9×10^{-6}	10^{-5}
dJ/du_0 (implicit BE)	tight inner tolerances	1.4×10^{-8}	10^{-5}
dJ/dn (implicit BE)	tight inner tolerances	5.3×10^{-6}	10^{-5}

Table 1: Relative errors of analytic derivatives against central finite differences of the full nonlinear computation. Every check in this table is an automated test in RDycore’s `cctest` suite, run on every change to the repository.

Parameter Jacobian. Manning n appears only in the drag term, so $\partial S_{\mathbf{q}}/\partial n = -2gnh^{-7/3}\mathbf{q}\|\mathbf{q}\|$: two nonzeros per cell, supplied to `TSSetRHSJacobianP`.

Verification. Three layers, all against central finite differences, with the pass/fail tolerances fixed in advance of the runs (Table 1): an edge-level harness that differences the flux routine itself on sampled states (with a Richardson self-check of the FD order); full-matrix comparisons of the assembled Jacobian against a matrix-coloring FD Jacobian of the actual RHS on three state/boundary combinations, boundary rows included; and end-to-end gradient checks through the adjoint (Section 2.2.2). A finite-difference coloring Jacobian (`numerics.jacobian: fd`) remains available permanently as the verification baseline.

2.2.2 Adjoint sensitivities

For gauge observations y_k at times t_k with observation operator H (one row per gauge, selecting water height) and error variance σ^2 ,

$$J(n) = \frac{1}{2\sigma^2} \sum_k \|Hu(t_k) - y_k\|^2. \quad (4)$$

`TSAdjoint` [19] differentiates the discrete time stepper, so gradients are exact to roundoff for the computed trajectory, which is the property that makes the FD gates in Table 1 meaningful. Multi-time observations are handled by a segmented backward sweep: the adjoint is advanced between observation times with `TSAdjointSetSteps` and receives a jump $H^T \sigma^{-2} r_k$ at each t_k (Algorithm 1 in Section 2.3). One sweep yields both dJ/du_0 (state sensitivity) and dJ/dn (per-cell parameter gradient, via the two-nonzero parameter Jacobian).

Two integrator facts constrain the design. PETSc’s plain forward-Euler integrator implements no adjoint, so adjoint-path configurations use the Runge–Kutta family. And for implicit integrators the discrete-adjoint gradient is only as exact as the inner solves: the Manning gradient fails its gate at default inner tolerances and passes once they are tightened (Table 1, implicit rows). We found loose inner tolerances a false economy on this problem: a tighter Krylov tolerance makes the implicit forward *faster*, because better linear solves let Newton converge in fewer iterations, and the production setting (Krylov 10^{-4} , nonlinear 10^{-5}) reproduces the fully converged answer to every printed digit for about 25% more wall time than the loosest usable setting. Campaigns therefore

run at converged inner tolerances, and the verified gradients *are* the production gradients. The FD gates cannot certify the objective’s own nonsmoothness over long windows, which Section 6.1 quantifies.

2.3 The calibration algorithm: strong-constraint 4D-Var

The complete calibration loop is summarized in Algorithms 1–3. It is a strong-constraint 4D-Var iteration: the model is enforced exactly (the only controls are the Manning field and, optionally, the initial state). Each objective evaluation performs one checkpointed forward sweep that records the gauge residuals r_k at the observation times, followed by one backward sweep of the discrete adjoint (Algorithm 2). The per-iteration cost is therefore one forward sweep, the checkpoint recomputation the backward sweep requires, and one adjoint sweep, about 2.1 forward solves in total at production settings (Eq. (5)) and independent of the number of parameters. We measure that constant rather than assume it, since for a nonlinear model on a memory-limited device the checkpointing scheme, not the adjoint, sets it. Throughout, the objective splits as $J = J_{\text{mis}} + J_{\text{prior}}$, where the *misfit* $J_{\text{mis}} = (2\sigma^2)^{-1} \sum_k \|r_k\|^2$ measures disagreement with the observations and J_{prior} is the Tikhonov term of Eq. (6). Only J_{mis} needs the adjoint; the prior term’s gradient is closed form.

The backward sweep carries two vectors with distinct roles. The *adjoint state* λ holds $\partial J_{\text{mis}}/\partial u(t)$ at the current point of the backward traversal. It is seeded with the terminal observation sensitivity $\sigma^{-2}H^T r_K$, propagated one step at a time by the transposed Jacobian of the forward step map (the backward “model step”), and incremented by a jump $\sigma^{-2}H^T r_k$ each time the sweep crosses an observation time, because the state at that time enters the misfit directly. The *parameter sensitivity* μ starts at zero and accumulates, at every step, the chain-rule contribution $(\partial\Phi_s/\partial n)^T \lambda$ of that step’s Manning dependence. When the sweep reaches $t = 0$, μ is the complete misfit gradient dJ_{mis}/dn and λ is dJ_{mis}/du_0 . Both transposed Jacobians are the analytic operators of Sections 2.2.1–2.2.2, the assembled RHS Jacobian for the state propagation and the two-nonzero parameter Jacobian for the μ accumulation, applied within TSAdjoint’s discrete adjoint of the chosen integrator.

The outer update is TAO’s bound-constrained limited-memory quasi-Newton method BLMVM [20] (Algorithm 3). The loop terminates on the standard bound-constrained test: the projected gradient g^P (the gradient with components pointing out of the feasible box zeroed) drops below an absolute tolerance, or an iteration budget is reached. The same loop serves every parameterization in this paper: per-region, per-cell, and land-cover-class fields differ only in the expansion of the parameter vector onto cells (line 3 of Algorithm 1) and the corresponding summation of μ back onto parameters.

What a gradient actually costs. The textbook statement that an adjoint gradient costs “about one extra forward solve” assumes the forward trajectory is available to the backward sweep. For a nonlinear model it is not: the trajectory must be stored or recomputed, and at basin scale on GPUs it cannot be stored. Checkpoint recomputation is therefore a real and unavoidable term, and we measure it rather than assert it. On a one-simulated-hour window of the 2.93M-cell mesh (3,600 implicit steps at $\Delta t = 1$ s, 400 in-memory checkpoints, four A100 nodes) a forward step costs 62.9 ms and an adjoint step 14.8 ms, and the backward sweep triggers 3,199 revolve recomputation steps, 0.89 of an additional forward. One objective-plus-gradient evaluation is thus

$$\underbrace{1.00}_{\text{forward}} + \underbrace{0.89}_{\text{recomputation}} + \underbrace{0.24}_{\text{adjoint sweep}} \approx 2.1 \text{ forward solves}, \quad (5)$$

about nine minutes at four nodes. Note where the cost sits: the adjoint sweep itself is the *cheapest* of the three terms, and recomputation is the expensive one, so the ratio is a property of the checkpoint

Algorithm 1 Strong-constraint 4D-Var calibration of the Manning field, as implemented with TAO/BLMVM and TSAdjoint. The window map $M_{t_{k-1} \rightarrow t_k}$ is the composition of m steps of the discrete forward stepper (explicit RK, backward Euler, or ARK-IMEX).

Require: prior/initial guess n_0 , bounds $[n_{\min}, n_{\max}]$, observations y_k at times $t_k = k m \Delta t$ ($k = 1, \dots, K$), gauge operator H , error σ , Tikhonov weight β , gradient tolerance ε_g , iteration budget $I_{\max}$

- 1: $n \leftarrow n_0$
- 2: **repeat**
- 3: set the model's cellwise drag coefficient from $n \triangleright$ identity for per-cell n ; constant over each region or land-cover class for the reduced parameterizations
- 4: $u \leftarrow u_0$ $\triangleright$ fixed initial state (strong constraint)
- 5: **for** $k = 1, \dots, K$ **do** $\triangleright$ forward sweep; trajectory checkpointed — every step, or a revolve schedule for long windows (Section 4)
- 6: $u \leftarrow M_{t_{k-1} \rightarrow t_k}(u; n)$
- 7: $r_k \leftarrow Hu - y_k$
- 8: **end for**
- 9: $J \leftarrow \frac{1}{2\sigma^2} \sum_{k=1}^K \|r_k\|^2 + \frac{\beta}{2} \|n - n_0\|^2$
- 10: $(\lambda, \mu) \leftarrow \text{ADJOINTSWEEP}(\{r_k\})$ $\triangleright$ Algorithm 2 differentiates the stepper above, so $\mu = dJ_{\text{mis}}/dn$ is exact to roundoff and inner-solver tolerance (Section 2.2.2)
- 11: $g \leftarrow \mu + \beta(n - n_0)$
- 12: $n \leftarrow \text{BLMVMSTEP}(n, J, g)$ $\triangleright$ Algorithm 3
- 13: **until** $\|g^P\| \leq \varepsilon_g$ or $I_{\max}$ iterations
- 14: **return** n

Algorithm 2 ADJOINTSWEEP: segmented discrete-adjoint backward sweep (TSAdjointSolve with TSAdjointSetSteps). Φ_s denotes the forward map of step s , $u_s = \Phi_s(u_{s-1}; n)$; its transposed Jacobians are built from the assembled RHS Jacobian $\partial f/\partial u$ (Section 2.2.1) and the parameter Jacobian $\partial f/\partial n$ according to the integrator's discrete-adjoint formula.

Require: checkpointed trajectory $\{u_s\}_{s=0}^{Km}$, residuals $\{r_k\}_{k=1}^K$ at steps $s = km$, gauge operator H , error σ ; as in Algorithm 1, m is the number of steps between consecutive observation times, so the window is Km steps long

Ensure: $\lambda = dJ_{\text{mis}}/du_0$, $\mu = dJ_{\text{mis}}/dn$

- 1: $\lambda \leftarrow \sigma^{-2} H^T r_K$ $\triangleright \lambda$ holds $\partial J_{\text{mis}}/\partial u(t_s)$ throughout
- 2: $\mu \leftarrow 0$
- 3: **for** $s = Km, \dots, 1$ **do**
- 4: restore u_{s-1} (and stage values): read the checkpoint, or recompute forward from the nearest one when the checkpoint schedule demands
- 5: $\mu \leftarrow \mu + (\partial \Phi_s / \partial n)^T \lambda$ $\triangleright$ accumulate this step's Manning dependence
- 6: $\lambda \leftarrow (\partial \Phi_s / \partial u_{s-1})^T \lambda$ $\triangleright$ backward model step (transposed step Jacobian)
- 7: **if** $s - 1 = km$ for some $1 \leq k < K$ **then**
- 8: $\lambda \leftarrow \lambda + \sigma^{-2} H^T r_k$ $\triangleright$ observation jump: u_{km} enters J_{mis} directly
- 9: **end if**
- 10: **end for**
- 11: **return** (λ, μ)

Algorithm 3 BLMVMSTEP: one iteration of TAO’s bound-constrained limited-memory variable-metric method [20]. $P_{[n_{\min}, n_{\max}]}$ is componentwise clipping to the bound box; the curvature pairs $\{(s_i, y_i)\}$ persist across calls. The projected gradient used in the convergence test of Algorithm 1 is $g_i^P = g_i$ for n_i strictly inside the box, $\min(g_i, 0)$ at the lower bound, and $\max(g_i, 0)$ at the upper bound.

Require: iterate n , objective J , gradient g , stored pairs $\{(s_i, y_i)\}$, bounds $[n_{\min}, n_{\max}]$

- 1: $d \leftarrow -B g^P \quad \triangleright B \approx (\nabla^2 J)^{-1}$ by the L-BFGS two-loop recursion over the stored pairs
- 2: find a step length $\tau > 0$ with a projected line search on $n(\tau) = P_{[n_{\min}, n_{\max}]}(n + \tau d)$ satisfying sufficient decrease of $J \quad \triangleright$ each trial τ costs one forward-plus-adjoint evaluation (lines 3–11 of Algorithm 1)
- 3: $n^+ \leftarrow n(\tau)$; evaluate g^+ at n^+
- 4: $s \leftarrow n^+ - n$; $y \leftarrow g^+ - g$; store (s, y) if $s^T y > 0$, discarding the oldest pair beyond the memory limit
- 5: **return** n^+

budget rather than of the adjoint. More checkpoints buy less recomputation until device memory runs out, and the 400 used here are near optimal for this step count. What does *not* change with the parameter count is the whole of Eq. (5): the same 2.1 forwards deliver a gradient with respect to one parameter or to 2.93 million, which is the property that makes distributed calibration tractable.

Each trial step of the BLMVM line search costs one such evaluation. A basin-scale window needs more iterations than a single batch allocation provides, so a run writes its parameter vector on exit and the next reads it back as its starting iterate, matched by land-cover class rather than by position. The resume is exact in the parameters (the continuing run’s first objective reproduces the previous run’s last to six digits), but BLMVM’s stored curvature pairs do not survive it, so a chain of jobs is not step-for-step identical to one long run. On a twin at the Houston mesh we found the restart was not a penalty: a chained 2 + 2 iteration run reached a lower objective than a single cold four-iteration run, the restart acting as a rescaling of the quasi-Newton model. That is one twin, not a rule, and the point for planning is that chaining did not cost convergence there.

Scaling the calibration problem. A quasi-Newton method needs its variables and its objective to be $O(1)$, and in physical Manning units this problem is not: on the production window $\|g\| \approx 1.5 \times 10^3$ while $n \approx 0.05$. Starting from $B = I$, the first trial point $n - g$ therefore projects every component onto the lower bound, where the model cannot solve (a uniform $\alpha = 0.2$ diverges, Table 10), so every line-search trial fails and the optimizer returns having taken no step. The failure presents as a solver defect: nothing is wrong with the gradient, and the finite-difference gates pass throughout.

Two changes make the problem well scaled, and both have physical readings rather than being tuning constants. First, optimize the *relative* field $\alpha_k = n_k/n_{\text{prior},k}$ and the relative objective $J/J(n_{\text{prior}})$. Here and throughout, α is a multiplier on roughness, never a step length: a class at $\alpha = 1.5$ carries 1.5 times its lookup value, and the uniform scans of Section 6.2 set every class to the same α . Both are then $O(1)$ and the unit quasi-Newton step becomes a roughness change of order 10%: at the NLCD start point of the production configuration, over a shortened window that reproduces the zero-step failure exactly, $\|g\|/J_0 = 0.10$, and the same start point that took no step in physical units takes one that lowers the objective and the peak-WSE MAE together. The bounds become statements about relative roughness, $\alpha \in [0.3, 3]$, which additionally excludes the region where the implicit solve fails. Second, set the Tikhonov weight from a prior standard

deviation on Manning n rather than by hand,

$$J = \underbrace{\frac{1}{2\sigma^2} \sum_k \|r_k\|^2}_{J_{\text{mis}}} + \underbrace{\frac{\beta}{2} \|n - n_{\text{prior}}\|^2}_{J_{\text{prior}}}, \quad \beta = \sigma_n^{-2}, \quad (6)$$

and report σ_n . This matters because β is dimensional: a bare value is uninterpretable across problems, and ours had been chosen independently in every experiment, spanning six orders of magnitude. Against the measured $J(\alpha)$ curve, $\sigma_n = 0.015$ places the minimum of the uniform-scaling direction at $\alpha \approx 0.70$, while $\sigma_n = 0.020$ leaves no interior minimum there at all, the same pathology the hand-chosen weight had. We ran the scan point at $\alpha = 0.70$ afterwards and measured an MAE of 0.6894 m against the 0.6895 m the balance implied (Table 10). Stating the prior turns the choice into an arithmetic consequence of a stated uncertainty on Manning n .

Absolute or fractional prior. Equation (6) states the prior as an absolute departure in n , and for a land-cover table that is the wrong statement. A single $\sigma_n = 0.015$ over an NLCD lookup spanning $n = 0.027$ (barren) to $n = 0.16$ (developed high) asserts a $\pm 56\%$ prior on one class and $\pm 9\%$ on another. Because the penalty scales as n_{prior}^2 , a small- n class buys a large *fractional* excursion cheaply. The iterates show that signature: at the production configuration’s first quasi-Newton step, every class driven to the upper bound was small- n (pasture 0.038, developed-open 0.040, developed-low 0.090) and every class driven to the lower bound was large- n (developed-high 0.160, developed-medium 0.120, shrub 0.115, woody wetland 0.098), an ordering by prior value, not by hydrology. Penalizing the fractional departure $n_k/n_{\text{prior},k} - 1$ with a width σ_α instead treats the classes alike, and states the prior as the percentage by which the lookup can be wrong.

The tension survives the reformulation, and that is the substantive result rather than an implementation detail. We ran the fifteen-class calibration under the fractional prior at the width we assign to the lookup, $\sigma_\alpha = 0.30$. Its first quasi-Newton step drove seven of the fifteen classes to a bound (developed-open, developed-low and pasture to $\alpha = 3$; developed-medium, developed-high, shrub and woody wetland to $\alpha = 0.3$), and after three iterations developed-medium and shrub were still on the floor ($n = 0.036$ and 0.0345 , below the lookup’s 0.040 for developed open land) with developed-low at twice its lookup value, for a peak-WSE MAE of 0.6154 m (Table 12) against the 0.6116 m the absolute prior reached in nine. A wider prior reaches the same fit sooner, by the same route, and the route is the problem. *At that width, this observable cannot deliver a fifteen-class roughness field inside the prior.* What it can deliver is the subject of Sections 6.3 and 6.4: the marks determine about three roughness combinations at this width, and a calibration confined to those converges and reaches the same answer from every prior we ran, with two of the three classes on the same $\alpha = 0.3$ floor. We report the prior width used with every number; its value is a judgment about the land-cover lookup, not a tuning parameter.

The failure mode is not specific to this model: an optimizer that takes zero steps on a problem whose gradient passes every verification gate is a scaling defect, and the cheapest test is to compare $\|g\|$ against the scale of the variable before suspecting the derivative.

3 Model configurations

The results that follow run the machinery of Section 2 on three domains, at resolutions two orders of magnitude apart and against four different observables. Because the paper’s central claim is comparative (what a parameter can explain *on a given observable and window*), the configuration

Configuration	Mesh	Window	Observable	Parameters
Verification twin	planar, single region	1.8 s, 40 steps	water height	per-cell, per-region
Houston twin	1 km, 2,746 cells	4200 s, $\Delta t = 30$ s	343 or 686 gauges, 14 times	2,746 per-cell
Houston, real network	1 km, 2,746 cells	4200 s, $\Delta t = 30$ s	17 USGS gauges, 238 obs	2,746 per-cell
Turning, class twin	30 m, 2.93M cells	early transient	13 gauges, 260 obs	15 NLCD classes
Turning, dense twin	30 m, 2.93M cells	early transient	418,076 strided cells	15 NLCD classes
Turning, mark twin	30 m, 2.93M cells	1 h, rain-forced	108 marks, peak WSE	15 NLCD classes
Turning, full event	30 m, 2.93M cells	72 h, $\Delta t = 1$ s	108 marks, peak WSE	forward only
Turning, production	30 m, 2.93M cells	12 h, event h29–41	46 marks, peak WSE	global scale, 15 classes

Table 2: The configurations used in Sections 4–7. Every row above *Turning, full event* is a twin, observing model output generated from a roughness field we chose. The twins are load-bearing rather than preparatory: a recovered field can be scored only against a known truth, and that score is what separates *this observable cannot determine the parameter* from *the derivative is wrong*. The real survey admits no such check, so the twins are what license reading Sections 6–7 as statements about information rather than about correctness. The last five rows are the 30 m Hurricane Harvey domain, where the observable changes from stage gauges to surveyed high-water marks for the reason given in Section 6.1. Every number reported against the real survey (baseline, scans, calibration, and spectrum) is measured on the production row, so those results compose.

is part of every result rather than background to it. Table 2 collects the configurations so that later sections can name one instead of restating it.

The twins. The verification twin is a planar, 40-step configuration whose only purpose is to exercise the derivative machinery where the objective is smooth; it carries the finite-difference gates of Table 1 and the noise study of Table 4. The Houston twin is a 1 km Hurricane Harvey mesh of 2,746 cells in which every cell is wet and moving, so all 2,746 per-cell parameters are observable in principle; truth is a two-zone field ($n = 0.03$ west, $n = 0.06$ east of the domain midline) and the recovered field is scored against it. Its real-network variant replaces the synthetic gauges with the seventeen in-mesh USGS gauges carrying Harvey stage records.

The 30 m Harvey domain. The Turning mesh resolves the Houston area at 30 m in 2.93M cells, with one Manning value per NLCD land-cover class (15 classes) and the published NLCD lookup as both prior and, for twins, truth. Three configurations share it. The class and dense twins ask whether fifteen classes are identifiable from the real gauge geometry and from a saturating observation set respectively. The mark twin replaces stage with peak water-surface elevation at the 108 QC-passed high-water-mark cells. The production configuration is the one every real-survey number is measured on: a 12-hour window over the upstream crest cluster (event hours 29–41), initialized from an hourly checkpoint of the 72-hour forward with the rain re-aligned to the window start (both validated bit-exactly), scored against the 46 surveyed marks whose modelled crest falls inside it. Section 6.2 explains why the observable, not the budget, sets that window.

Shared solver settings. Every adjoint-path configuration uses the explicit source treatment (source.method: explicit) with stiffness handled by the implicit integrator rather than inside the right-hand side (Section 2.1), and runs at converged inner tolerances (Krylov 10^{-4} , nonlinear 10^{-5}) so that the verified gradients are the production gradients (Section 2.2.2). Calibrations use TAO/BLMVM with Manning bounds $[0.01, 0.2]$, and long windows use revolve checkpointing (Section 4).

4 Implicit stepping, enabled and required

The explicit drag term is stiff in thin water: linearizing gives a stability limit $\Delta t \lesssim 2h^{4/3}/(gn^2|v|)$, seconds for sheet flow at centimeter depths, against the 30 s gravity-wave step of the 1 km mesh. On the Houston 1 km configuration ($\Delta t = 30$ s) we found the explicit source treatment returns a non-finite state within the first objective evaluation, as the estimate predicts. The same holds at basin scale, where the margin is a factor of 400: on the spun-up 30 m, 2.93M-cell Harvey mesh (99.9% wet, median depth 6 mm) the friction-rate limit is $\Delta t \lesssim 2.4$ ms, and forward Euler of the differentiable right-hand side returns a non-finite state at every tested step ($\Delta t = 0.25, 0.125, 0.05$ s) while backward Euler at $\Delta t = 1$ s completes a 72-hour window without a Newton failure. There is no viable explicit path for the differentiable right-hand side at this resolution.

The assembled exact Jacobian supplies the remedy directly: backward Euler via PETSc’s theta method, with SNES driven by the same Jacobian, carries the Harvey window at $\Delta t = 30$ s, and the theta-method adjoint passes the same FD gates (Table 1).

We also implemented the sharper tool: an ARK-IMEX splitting with the Manning drag as the stiff implicit part (`temporal: arkimex`). The implicit stage system is block-diagonal (an independent 3×3 Newton per cell, no global solve) while fluxes and bed slope remain explicit under the gravity-wave CFL limit. Its adjoint passes the same gates (2.0×10^{-8} for dJ/du_0 , 2.5×10^{-6} for dJ/dn) and it carries the Harvey window at $\Delta t = 30$ s at 6.7 s per forward-plus-adjoint sweep, with gradients matching backward Euler to three digits. The splitting also runs under RDycore’s libCEED operator backend, whose explicit side uses friction-free source Q-functions; the two backends agree to machine precision (5×10^{-16} relative) after 100 ARK-IMEX steps. A device-resident implicit part is the natural follow-up.

Three implementation facts apply to any adjoint retrofit on PETSc’s time steppers:

- PETSc’s ARKIMEX adjoint dereferences *both* the implicit- and explicit-part parameter Jacobians, so a zero explicit-part `TSSetRHSJacobianP` must be registered even when the parameter appears only implicitly.
- The embedded step-size controller must be disabled for finite-difference gradient validation: perturbed solves otherwise take different step sequences, which shows up as an apparent $\sim 10^{-4}$ gradient error that vanishes with fixed steps.
- PETSc’s in-memory checkpointing trajectory cannot tolerate a *segmented* forward solve (one `TSSolve` per observation window). `TSSolve` sets its converged reason before the final trajectory write of every call, which the memory trajectory interprets as the end of the forward run, corrupting the checkpoint schedule at every interior observation time. The forward sweep must be a single `TSSolve` with observations recorded by a monitor; the segmented backward sweep is unaffected.

Newton robustness is set by residual discontinuities, not by conditioning. Where implicit stepping fails on this problem, it fails for a reason that is invisible in the smooth twin experiments: the residual is discontinuous, so Newton has nothing to converge *to*. At the land-cover roughness prior, rather than the twins’ smooth low- n fields, backward Euler descends for a dozen iterations and then stalls: a line-search step of negligible length changes the residual norm severalfold, which is the signature of a jump, not of ill-conditioning. Two mechanisms are responsible, both invisible until the parameter field is realistic: the drag’s wet/dry cutoff, whose amplitude scales as n^2 , and the critical-outflow boundary’s stagnation switch. Both are discontinuities of the residual rather than of its derivative, and both are removed by changing the model rather than

Rain-forced configuration (land-cover prior)	Outlet	Outcome
600-step control	critical outflow	Newton pinned from step 498
600-step control	free outflow	zero failures; 598/600 solves in two iterations
1-h calibration window (forward + adjoint + line search)	free outflow	clean end to end
72-h forward through the Harvey crest (2.93M cells)	free outflow	259,200 solves, zero failures

Table 3: The outlet replacement is a cure, not an improvement. Under critical outflow, tolerance relief bought steps only linearly; the transmissive outlet completes the identical window and every longer one attempted.

the solver. The drag $gn^2h^{-7/3}\mathbf{q}\|\mathbf{q}\|$, gated at the dry cutoff with a plain $\mathbf{q}/h$ velocity, switches an $O(h^{-7/3})$ term on and off as a cell crosses the cutoff; evaluating it with the regularized velocity $\mathbf{u} = \mathbf{q}h/(h^2 + h_{\text{anuga}}^2)$ of ANUGA, with $h_{\text{anuga}} = 0.001$ m, makes the drag vanish as $h^{5/3}$ instead and reduces to the plain form at $h_{\text{anuga}} = 0$. The value is the smallest that removes the stall: 0.003 m is indistinguishable from it and 0.01 m is unstable. The outflow condition is replaced in the same spirit.

Both fixes are implemented, gated, and opt-in. Regularizing the drag velocity removes the first and restores convergence over the early transient. The second we removed by replacing the outlet condition itself: a transmissive (zero-gradient) free outflow, in which the ghost state copies the interior state (the convention of most finite-volume shallow-water codes), is continuous in the interior state by construction, and its ghost map is the identity, so the analytic Jacobian and adjoint carry it exactly, FD-gated like every other block. The effect is not an improvement but a cure (Table 3): the identical failing control completes with zero nonlinear failures, and the fix holds through the longest windows we run. Before the replacement we absorbed the stall by loosening the nonlinear tolerance, and a class-twin calibration under that setting still recovered all fifteen class values to machine precision, so the gradient path was unimpaired. The general lesson for adjoint retrofits is that nonsmoothness in an operational flood code is a nonlinear-solver question as well as a differentiability question.

Checkpointed adjoints for long windows. A stored trajectory for a full 36-hour Hurricane Harvey window at the 30 m mesh’s CFL-limited $\Delta t = 0.25$ s (518,400 steps, 2.93M cells) would occupy tens of petabytes. PETSc’s checkpointing trajectory (`-ts_trajectory_type memory` with the revolve optimal-schedule library) replaces storage with recomputation. We validated it across the full hierarchy. On the dam-break twin it reproduces the disk-trajectory calibration and gradient gates bit-for-bit for explicit RK, backward Euler, and ARK-IMEX (the implicit integrators need the same tight inner tolerances as in Section 2.2.2, since checkpoint recomputation re-solves the stage systems). On the 30 m mesh, a forward-plus-adjoint sweep with only *three* states checkpointed in memory matches the disk-trajectory gradient to all printed digits, and runs faster (recomputation is cheaper than trajectory I/O at this scale). For the full window, a few hundred checkpoints (~ 14 GB aggregate) bound the recompute overhead by a factor of about four, making the long-window adjoint memory-feasible; the remaining cost is the forward compute itself, measured in Eq. (5).

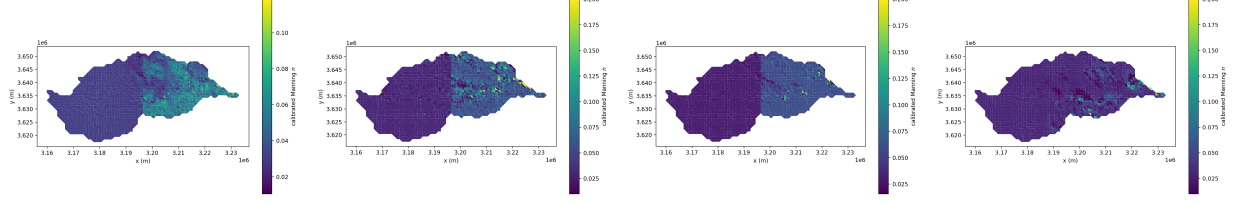


Figure 1: Regularization, semiconvergence, and network density on the Houston twin (2,746 parameters; truth: $n = 0.03$ west, $n = 0.06$ east of the midline; implicit stepping, 14 observation times). Far left: weak regularization ($\beta = 10^{-5}$, 343 gauges) stopped early at 300 iterations — 26.3% error. Center left: the same setup run to 2000 iterations — the misfit is lower but the field drifts along unobserved directions to 38.0% error and saturates the bounds $[0.01, 0.2]$. Center right: proper Tikhonov weight ($\beta = 10^{-4}$, 686 gauges), 1000 iterations — 19.4% error, no drift, sharp zone boundary. Far right: the *real* network — the 17 in-mesh USGS gauges with Harvey stage records, same $\beta = 10^{-4}$, 1000 iterations — the misfit falls 2×10^7 -fold while the field recovers to only 44.9%: no trace of the two-zone truth, near-prior almost everywhere with null-space structure pinned against both bounds.

5 Twin experiments: what the observing network can recover

In a twin experiment the observations are model output generated from a roughness field we chose, so a calibration can be scored against a known truth rather than against a real survey whose own errors are unknown. The per-cell twin runs on the Houston 1 km Hurricane Harvey mesh (2,746 cells, 4200 s window at $\Delta t = 30$ s). Truth is a two-zone field split at the domain midline, observed at 343 synthetic gauges over 14 observation times, calibrated with implicit stepping and Tikhonov regularization ($\beta = 10^{-5}$). Every cell is wet and moving in this configuration, so all 2,746 parameters are observable. The configuration is deliberately gauge-sparse (343 gauges for 2,746 parameters, an 8:1 ratio) and it exhibits the classic *semiconvergence* of underdetermined inverse problems (Figure 1). At 300 BLMVM iterations (~ 22 minutes on one laptop core, ~ 4.4 s per forward-plus-adjoint) the recovered map shows the two-zone truth structure at 26.3% relative L^2 error. Continuing the same weakly regularized optimization to 2000 iterations *degrades* recovery to 38.0%: the misfit keeps decreasing while the parameters drift along observation-null directions, ultimately pinning cells against both bound constraints (Figure 1, center left). Early stopping was acting as the effective regularizer. The behavior disappears when the problem is made well posed: a properly regularized Houston run ($\beta = 10^{-4}$, 686 gauges) reaches 19.4% at a 1000-iteration budget with no drift (Figure 1, center right), the sharpest recovery of the three, with the zone boundary resolved at the truth midline. This is the overfitting risk documented for distributed friction estimation [8], reproduced in a controlled twin. It motivates the regularization studies (total variation, land-cover priors, discrepancy-principle stopping) planned as follow-on work.

The real gauge network. The synthetic gauge sets above are a controlled ramp; the network that matters is the real one. Twenty USGS gauges fall inside the Houston mesh, seventeen with Hurricane Harvey stage records.¹ In the same twin with the observation operator built at those seventeen gauge cells (238 observations, 17 gauges $\times$ 14 times, against 2,746 parameters), BLMVM drives the misfit down by a factor of 2×10^7 (the synthetic observations carry no noise and are fit

¹HydroShare DOI 10.4211/hs.c037167e497546a1bc1508dfb32a9cff.

to roundoff) while the recovered field is 44.9% from the truth and saturates both bounds (Figure 1, far right). This is semiconvergence in its purest form: the real network’s null space is so large that data fit says almost nothing about parameter recovery, and no Tikhonov weight or stopping rule rescues a per-cell field from 238 observations. The practical conclusion for real-data calibration is to shrink the parameter space to match the network. Shrinking it is necessary and not sufficient, as the next subsection measures: whether a reduced parameterization is identifiable depends on the observation geometry and window, not on the ratio of observation to parameter counts.

5.1 Identifiability is a property of the observable: gauges versus high-water marks at 30 m

The natural reduced parameterization for this domain is one Manning value per NLCD land-cover class (15 classes at the 30 m Turning mesh, 2.93M cells), with the published NLCD lookup as prior and truth for twins. Counting favours it: even the sparse real network supplies an order of magnitude more observations than parameters. Measurement says otherwise. In a class twin on the 30 m mesh at the real 13-gauge geometry (260 observations, a short early-transient window), the objective falls $128\times$ while the recovered class values remain 66% wrong in relative L^2 , worst class 81%. The failure is structural: classes whose cells are dynamically distant from every gauge (the three forest classes) never move from the starting value, because within the window nothing their roughness does reaches a gauge. The identical configuration observed at 418,076 strided cells recovers all fifteen classes to machine precision, so the failure is the network, not the method. Fifteen parameters and thirteen gauges, an “overdetermined” observation count, remain unidentifiable.

The same question re-asked with the peak observable comes out differently. Observing the same NLCD-truth twin at the 108 QC-passed high-water-mark cells (peak WSE sampled over a one-hour rain-forced window, implicit stepping, free-outflow outlet; Section 6.1) and calibrating fifteen classes from a uniform start: the objective falls $81\times$, the peak-WSE error at the marks drops from 0.103 m to 0.017 m MAE, and the area-dominant classes are identified: the three developed-intensity classes, pasture, and crops recover to 0.7–10.5%, together 83% of the domain’s cells, for an area-weighted relative L^2 error of 0.20 (unweighted 0.58). The residual errors concentrate in small-area classes (forests, shrub, emergent wetland, $\sim 5\%$ of cells), which now *move* (they are sensitive, unlike in the gauge experiment) but overshoot to the bounds along weakly constrained directions. That is ordinary semiconvergence, treatable with regularization and longer windows, rather than a structural absence of signal. Three caveats apply: the one-hour window samples early-transient peaks, not flood peaks (the real calibration windows are 6–12 hours and should only help); the run was capped at 15 quasi-Newton iterations; and twin marks synthesized from the truth trajectory carry no representation error, so this is an identifiability statement, an upper bound on real-data performance rather than a forecast of it.

Noise turns identifiability into a signal-to-noise question. Every recovery above is noise-free, as are the verification twins behind Section 2.3. Such experiments establish that the gradient is right and that the observable carries the information; they do not establish that a field can be calibrated from real data. What separates the two is the size of the parameter-induced signal relative to observation error, and that ratio moves by orders of magnitude with the configuration (Table 4). On a short verification window the entire Manning-induced observation signal is a few microns of water height, so a noise-free optimizer descends on it indefinitely while a tenth of a millimetre of observation noise destroys the recovery outright, and no Tikhonov weight in a sweep up to $\beta = 1$ rescues it. At basin scale the same quantity is ~ 0.1 m of peak water-surface elevation after one simulated hour, five orders of magnitude larger, which is the regime in which a parameter

Configuration	Signal	Observation noise	Outcome
Verification twin (planar, 40 steps, 1.8 s)	5 μ m water height	none 0.1 mm 1 mm	per-region recovery 4×10^{-7} per-cell recovery destroyed (0.20 $\rightarrow$ 1.8) per-region destroyed ($4 \times 10^{-7} \rightarrow 0.83$, both at bounds)
30 m basin, 1 h	~ 0.1 m	none	area-weighted rel. L^2 0.20; MAE 0.103 $\rightarrow$ 0.017 m
rain-forced, 108 marks	peak WSE	0.15 m (HWM grade)	misfit falls 9.4 $\times$, MAE 0.23 $\rightarrow$ 0.12 m — but 5 of 15 classes pin to bounds

Table 4: Parameter-induced observation signal versus observation error. The noise-free recoveries differ by seven orders of magnitude in the signal they descend on; only the basin-scale row is a fair test against survey-grade error (0.1–0.3 m for high-water marks). Noise is a deterministic, decomposition-independent perturbation of the twin observations, so the comparison is reproducible across rank counts.

signal can exceed survey-grade high-water-mark error. The lesson is not that the method is fragile: it is that friction must act long enough to imprint a signal above the noise floor, which makes window length a first-class calibration decision rather than a cost parameter. The basin-scale noisy row shows what marginal signal-to-noise looks like in practice, and why misfit at the marks is the objective but not the result. With 0.15 m of observation noise, even the *true* field cannot beat a mark MAE of about $\sigma\sqrt{2/\pi} \approx 0.12$ m against the noisy values, and the optimizer reaches that floor while driving five of the fifteen classes onto their bounds (pasture, 428k cells, to the ceiling) and ending far from the true field it was supposed to recover. The data fit is as good as truth could achieve; the parameters that achieve it are wrong, because the last of the misfit was removed by matching noise. On real data the parameter error is invisible and the bound-pinning is its symptom, which is the argument for stronger regularization, a longer window, and discrepancy-principle stopping (stop at the noise level rather than descend past it) in the production run.

6 The real survey: what roughness can explain

The twins establish what the machinery can recover under controlled conditions; this section turns it on the real Hurricane Harvey survey. The observable changes first — the stage gauges are unusable at this resolution — and everything after that is measured against surveyed high-water marks: the uncalibrated baseline and the drainage defect it exposes (Section 6.1), how much of the remaining error roughness can remove (Section 6.2), the calibration (Section 6.3), and the spectrum that counts how many parameters the marks can actually determine (Section 6.4). Section 7 then points the same measurement at the initial condition.

6.1 High-water marks and the uncalibrated baseline

Calibration against the actual stage records is blocked by geometry, not software: at all 13 rain-driven USGS gauges in the 30 m domain, modelled water-surface elevation sits 1–11 m *above* observed stage, because a 30 m cell stores one mean ground elevation — the bank of an incised bayou — while the gauge hangs in the channel below it; the offset is time-varying as the real channel fills, so no per-gauge datum correction can repair it. Comparing the mesh directly against the stage record makes the geometry explicit (Table 6): over the calibration window the cell bed sits above the *peak* observed water surface at seven of the twelve gauges with data, and above the

Manning field	22	23	90	gauge J	mark MAE
NLCD lookup	0.090	0.120	0.098	31,313	0.7188
calibrated on the gauges	0.27	0.36	0.171	24,007	0.7609
calibrated on the marks	0.210	0.036	0.029	30,333	0.6290

Table 5: The same three classes (developed-low 22, developed-medium 23, woody wetland 90) calibrated against each observable and scored against both, at $\sigma_\alpha = 0.30$ with the other twelve classes frozen at the lookup. Manning n in $\text{s m}^{-1/3}$, gauge J over 134 above-bed stage observations, mark MAE in metres over the 46 cluster-A marks. Each calibration improves its own observable and the gauge calibration degrades the other, past the uncalibrated prior. The bound is $\alpha \in [0.3, 3]$, so 0.27 and 0.36 are at it.

water surface for 71% of all gauge-hours. Only the two Buffalo Bayou main-stem gauges, where the channel is wider than a cell, and the two gauges in the Addicks and Barker reservoir pools hold their water for the whole window. The mesh resolves the major channel and not the tributaries, which is where nine of the thirteen gauges sit.

Calibrating on the gauges instead. The four gauges that hold their water for the whole window are enough to pose the calibration the other way round, and doing so tests whether a roughness field fitted to one observable predicts the other. We calibrated the same three classes on the above-bed gauge stage (134 observations, an observation counting only while the observed water surface is above the cell bed) and scored the result on the 46 marks. The answer is that it does not, and the failure is directional rather than marginal (Table 5). The gauge calibration raises all three classes — developed-low and developed-medium to the $3\times$ upper bound, woody wetland to $1.74\times$ — where the mark calibration puts the latter two on the $0.3\times$ floor. The two observables place the same class a factor of ten apart, each at an edge of the same prior. Scored on the marks, the gauge-calibrated field reaches 0.7609m against the uncalibrated prior’s 0.7188: *calibrating on the gauges makes the model worse at the marks than not calibrating at all*, while lowering the gauge misfit by 23%.

The mechanism is the geometry of the preceding paragraph. At a mark on the floodplain the modelled peak is too high, so the misfit gradient asks for less conveyance; at a gauge on a bank cell the modelled stage is too low by the representation offset, so it asks for more. Neither request is about friction. The gauge residual is the size of that offset rather than of a roughness error: at the NLCD prior the above-bed gauge misfit is 3.24m RMSE, and the best field we obtained against it leaves 2.84m, so roughness spans 12% of a discrepancy that a 30m mesh of an incised channel would produce with no friction error at all. A calibration reported against the gauges alone would show a 23% misfit reduction and a physically impossible field.

We report this as the paper’s validation result. It is negative, and it is the one the survey-only numbers of Section 6.3 need: a three-parameter field that removes 12.5% of the mark error does not transfer to a second observable in the same basin and window, so that 12.5% is a fit and not a skill. The two observables agree on one thing, that the land-cover table is not where this model’s error lives.

We therefore changed the observable to the USGS high-water-mark archive: surveyed *peak* water-surface elevations on structures and debris lines, i.e. on the floodplain surfaces a 30m cell does represent, and the benchmark currency of the flood-modeling literature (Inunda reports 0.67m MAE against Harvey high-water marks [10]). Of 2,364 Harvey marks, 324 fall in the Turning domain; applying to the marks the same QC that disqualified the gauges — is the surveyed elevation

Gauge	cell bed (m)	window WSE (m)	above bed
Buffalo Bayou nr Katy	33.27	35.56–36.05	100%
Buffalo Bayou at Houston	6.93	10.64–12.77	100%
Langham Ck nr Addicks	28.23	30.70–31.28	100%
Bear Ck nr Barker	34.73	34.96	100%
Buffalo Bayou nr Fulshear	31.47	31.23–31.56	29%
Langham Ck at W Little York	34.10	32.76–34.09	0%
S Mayde Ck	34.60	34.07–34.29	0%
Whiteoak Bayou at Alabonson	23.60	22.26–23.48	0%
Brickhouse Gully	20.40	16.24–19.64	0%
Cole Ck	25.30	20.95–22.51	0%
Little Whiteoak Bayou	14.50	10.65–13.37	0%
Whiteoak Bayou at Main St	11.80	9.66–11.33	0%
Whiteoak Bayou at Houston	15.03	no record in window	

Table 6: The 30m mesh against the stage record at the thirteen rain-driven USGS gauges over the calibration window (event hours 29–41). Stage at these gauges is published as water-surface elevation (WSE) in NAVD88, the datum of the mesh, so the cell’s mean bed elevation and the observed water surface compare directly. “Above bed” is the fraction of the window’s 15-minute records with the observed water surface above the cell bed. At seven gauges the bed sits above even the window’s peak stage: the cell holds the bank, and no roughness value places water in it at the elevation the gauge reads.

above the containing cell’s bed? — rejects 62% of them (they are channel surveys), leaving 108 marks of quality fair or better. The mismatch this QC addresses is the one Xu et al. measured on a 30m simulation of the same event: validated against 48 marks, their simulation had an RMSE of 0.9m with 25 of 48 within 25%, which they attribute to comparing a point survey with a cell average [21]. Two choices here reduce that exposure. The QC keeps only marks on surfaces a cell represents, and the misfit compares water-surface elevations rather than depths, so the difference between the ground at the point and the mean ground of the cell enters once, through the cell bed, and not a second time through the depth. The misfit compares each mark cell’s peak simulated WSE over the window, $\max_k Hu(t_k)$, against the surveyed value; its adjoint needs no new machinery, because each mark’s residual can be injected at its own argmax time in the segmented backward sweep of Algorithm 2, which is the exact gradient wherever the peak time is locally constant. The parameter gradient of the peak misfit passes the same central-difference gates as every other block on the smooth verification twin. At basin scale the naive check fails, and a set of instrumented measurements pins down why, in a way that validates the gradient rather than the check (Table 7). No mark’s argmax moves and none flips wet/dry across any probe, so the discrepancy lives below the observable; the evaluations are deterministic to nine digits and the FD value is stable across a decade of probe step, which excludes noise, curvature truncation, and sparse kinks. Decisively, the gap *accumulates with trajectory length*: at the shortest window the gate passes at the inner-solve floor with every piece of machinery engaged, and it grows three orders of magnitude by basin window lengths. A proportional defect in $\partial f/\partial n$ or the adjoint accumulation would err the same fraction at any window length; trajectory-accumulated branch crossings — the probe shifts which cells cross the depth cutoff at some step — predict exactly this profile. Two further checks close it: as the probe shrinks, the FD converges to the adjoint direction by direction — the largest single-class direction passes the 10^{-5} gate outright at the smallest probe — and the fifteen per-class gaps *sum* to the domain-wide gap, the additivity that independent per-cell branch crossings predict and

Window	Steps	FD-vs-adjoint gap, domain-wide direction
smooth verification twin	40	9×10^{-7} (gate 10^{-5} : pass)
basin, 1 min	60	9×10^{-5} (inner-solve floor: pass)
basin, 5 min	300	8×10^{-3}
basin, 15 min	900	2×10^{-2}
basin, 30 min	1800	2×10^{-1}
basin, 1 h	3600	7×10^{-2}

Table 7: The FD gate’s failure accumulates with trajectory length on identical basin, rain, marks, and converged tolerances — the signature of branch crossings along the trajectory, not of gradient error. The 30-minute and 1-hour rows sample peaks over different horizons, so strict monotonicity across them is not expected; single-class directions sit one to three orders lower at every window.

no plausible defect would arrange. The measured conclusion: the adjoint returns the exact one-branch derivative; over long windows the objective is dense with wet/dry branch surfaces, and a finite-step central difference converges to a locally *smoothed* slope that sits several percent from any one branch. The quasi-Newton iteration consumes one-branch derivatives of a piecewise-smooth objective — standard practice for max-type and switching systems — and its observed descent (Section 5.1) is consistent with that footing.

The remaining scientific choice is the calibration window, which for marks that record the event maximum must cover the flood peak.

The uncalibrated baseline against the real survey. Because the marks record the event maximum, the baseline evaluation must simulate through the crest: a 72-hour rain-forced forward from the available initial state (2017-08-26 18:00, reaching past the Buffalo Bayou crest), NLCD-prior Manning, free outflow, backward Euler at $\Delta t = 1$ s on the 2.93M-cell mesh — 259,200 implicit steps, *zero* Newton failures, two hours on one GPU node. The model wets all 108 marks, and the uncalibrated peak-WSE error decomposes into two distinct populations (Table 8; confirmed mark for mark at fully converged inner tolerances). The marks that crest inside the window carry the error a Manning field can plausibly address. The others never crest at all: their cells fill monotonically through the entire window — every one peaks at the final hour with zero recession, still rising while the first population is already draining — accumulating some ten metres of standing water whose late-window inflow arrives laterally, after the rain has ended. Those cells sit in the downstream Buffalo Bayou reach: the model cannot drain it, a conveyance/outlet defect that no roughness value can absorb and that would corrupt any calibration asked to fit it. The obvious suspicion — that we simply failed to give the domain an exit — does not survive inspection of the mesh. The domain is a delineated watershed rather than a rectangular cutout (35 of its 6,198 boundary nodes lie on the bounding box), so its unflagged perimeter is a catchment divide and a reflecting wall is the right condition there; and its single free-outflow side set sits at the perimeter’s lowest point, the only boundary nodes at $z \approx 0$ against a boundary median of 29.9 m. The exit is present and correctly placed. What the ponded reach then reveals is a limitation of the idealization itself: it holds water above the elevation of parts of its own divide, which a reflecting wall retains and the real event would have spilled.

Three measurements on the 72-hour forward make that concrete. The 37 never-cresting mark cells are not local depressions in the terrain: one of the 37 is lower than every edge-neighbour, against six of the 71 marks that crest normally. Rainfall is not what fills them: on the basin-mean MRMS series, simulation hours 60–72 carry 1.7% of the event’s rain and the hours beyond 72 carry

Marks	n	MAE	bias
crest inside the 72-h window	71	1.51 m	+1.20 m
never crest (reach cannot drain)	37	7.06 m	+7.06 m
all	108	3.41 m	+3.21 m

Table 8: The uncalibrated baseline against the 108 surveyed high-water marks, decomposed. The second population is model-high at every single mark and identifies a reach with no working exit; the first is the calibration target.

Group	n	mean lon.	bias	MAE
crest h29–41 (upstream)	46	−95.695	+0.30 m	0.72 m
crest h48–72 (middle)	25	−95.568	+2.85 m	2.96 m
never crest (downstream)	37	−95.440	+7.06 m	7.06 m

Table 9: The crest-timing groups are spatial bands, and the model-high bias grows monotonically downstream toward the reach with no working exit. The upstream band is the calibration target; the middle band is contaminated by the same defect at intermediate strength and cannot serve as held-out data.

none. What the model holds there is a lake. Its peak water surface across all 37 cells is 20.0 ± 1.1 m, over surveyed peaks that run from 6.2 to 20.2 m and beds from 0.3 to 18.2 m: the surveyed water surface slopes toward the outlet and the modelled one does not (the correlation of surveyed WSE with the model’s excess is -0.93 , and a flat pond at 20.3 m predicts that excess to 1.1 m RMS). The divide sets the level. A quarter of the perimeter’s 6,198 nodes, all in the domain’s eastern quarter, lie below 20 m, while the free-outflow side set is thirteen element edges wide: the real flood left across the low divide, and the model can leave only through the outlet, so it ponds to the height its walls impose. The calibration target is therefore the 71 admissible marks; the drainage defect is reported as a finding in its own right; and the remaining bundle (initial-condition, rainfall, mesh-elevation, and datum error) is what the calibration must be prevented from absorbing along with the roughness signal. The literature’s calibrated benchmark on this event is 0.67 m MAE [10].

The defect is a gradient, not a switch. Sorting the marks by crest time splits them into three groups, and those groups turn out to be three *spatial bands* whose bias increases monotonically downstream (Table 9). The never-crest population is the end of a continuum, not a separate pathology: the intermediate band already carries +2.85 m of the same model-high signature. Two consequences follow. First, a calibration window has to be placed on the upstream band, where the model is physically sound — crests there fall in event hours 29–41, and the 46 marks in that band are the target of every production number below. Second, the intermediate band is *not* clean held-out data, which rules out the obvious validation design; out-of-sample skill has to be tested by cross-validation within the upstream band instead.

6.2 How much of the error can roughness explain?

Before asking an optimizer to fit fifteen land-cover classes it is worth asking what any roughness field could achieve, and that question has an optimizer-independent answer. Scaling the entire NLCD field by a single factor α traverses the natural first direction of the problem — close to, though as Section 6.4 measures, not exactly, its most identifiable one; evaluating the peak-WSE misfit along it bounds what this paper calls the *authority* of a parameter over an observable: the amount of the

α	n range	J	MAE	dry
0.20	0.005–0.032	implicit solve diverges		
0.30	0.008–0.048	6.737×10^2	0.6392	1/46
0.45	0.012–0.072	7.038×10^2	0.6614	1/46
0.60	0.016–0.096	7.406×10^2	0.6776	1/46
0.70	0.019–0.112	7.655×10^2	0.6894	1/46
0.80	0.022–0.128	7.813×10^2	0.6991	0/46
1.00 (NLCD)	0.027–0.160	8.083×10^2	0.7188	0/46

Table 10: Uniform roughness scan on the production configuration (12-hour window, 46 cluster-A marks); every cell carries $n = \alpha n_{\text{NLCD}}$, so the n range column is the whole lookup table scaled by α . J falls monotonically toward lower roughness with no interior minimum: the entire uniform-roughness knob is worth 0.08 m against a 0.72 m error, and only at roughness 30% of published values, where n reaches 0.008 and smooth concrete is ≈ 0.012 . Within a defensible range it is worth about 3 cm ($\alpha = 0.70$, itself the optimum implied by the prior width of Section 2.3). The limit at $\alpha = 0.2$ is numerical, not drying — one mark stays dry throughout.

model–survey error that changing the parameter can remove, measured without reference to how, or whether, a calibration converges. The largest such amount over the parameter’s defensible range is the parameter’s *ceiling*. Both are lengths, in metres, and fractions of the uncalibrated error, and the paper uses the two words in that sense only.

We ran that scan on the production configuration: a 12-hour window over the upstream crest cluster (event hours 29–41), initialized from an hourly checkpoint of the 72-hour forward and with the rain re-aligned to the window start (both validated bit-exactly), scored against the 46 surveyed marks whose modelled crest falls inside the window. Each point is one forward with no adjoint (Table 10).

The window is set by the observable. A high-water mark records an event maximum, so it carries information about roughness only if its crest occurs inside the calibration window; before its crest it is an inequality, and after it a constant. The measured crest times decide the window rather than any budget: they are bimodal, clustering in event hours 29–42 and 60–72 with an empty gap between, so a window spanning both would spend most of its length integrating marks that had already peaked. The upstream cluster is also the population the model can legitimately be scored against at all, since the downstream marks sit in a reach it cannot drain (Section 6.1) — a defect no window length repairs. The 46 marks used throughout this paper are therefore those for which the observable means what it is meant to mean, and every number reported here — the uncalibrated baseline, the scans, and the calibration — is measured on that same window, so they compose — including the spectrum of Section 6.4.

The entire uniform-roughness knob is worth 0.08 m against a 0.72 m error (Figure 2) — about 11% — and it reaches that only at $\alpha = 0.3$, i.e. Manning values 30% of the published NLCD lookup, as low as $n = 0.008$ where smooth concrete is ≈ 0.012 . Over a physically defensible range ($\alpha \gtrsim 0.7$) the knob is worth about three centimetres. The curve is mildly concave with no interior minimum: J falls monotonically toward lower roughness until the implicit solve fails.

That number is corroborated from an independent direction. The identifiability twin of Section 5.1 measured the peak-WSE signal induced by a large class perturbation at ~ 0.1 m — the same authority scale, from a different experiment. So 85–90% of the model-versus-survey residual is not roughness at all: it is rainfall forcing, mesh elevation, datum, and representation error at

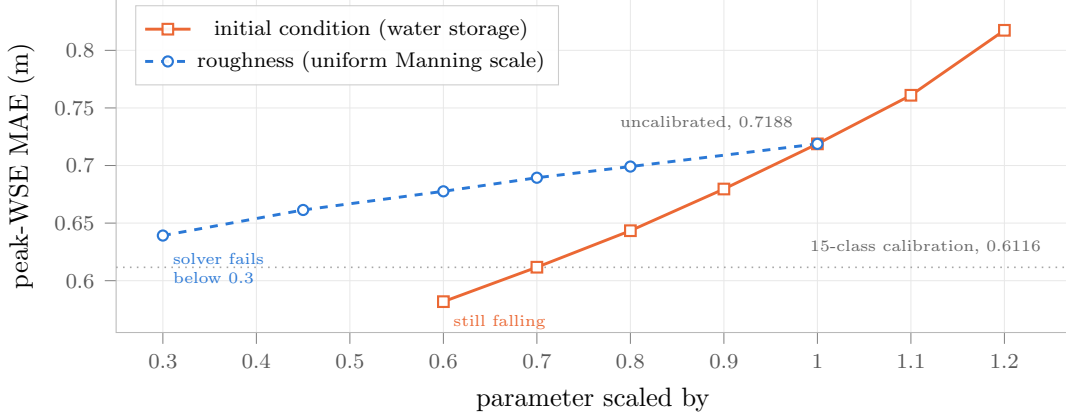


Figure 2: What each parameter can do to the peak-WSE error, on identical axes: the horizontal axis scales the parameter, the vertical axis is the same mean absolute error over the same 46 marks, and both curves pass through the uncalibrated model at 1.0. Roughness is concave and runs out — and its useful end is physically indefensible, reaching $n = 0.008$ before the implicit solve fails. The initial condition is about four times steeper per unit fractional change, convex, and has not turned over at a 40% reduction in antecedent water. The dotted line is the fifteen-class calibration, which lies off both curves because it redistributes roughness rather than scaling it.

Field	J	ΔJ	MAE	ΔMAE
NLCD prior	808.3	—	0.7188	—
developed classes 21–24 only (71.6% of cells)	786.8	−21.5	0.7306	+0.0118
everything else (28.4%)	827.5	+19.2	0.7268	+0.0080
both (= uniform $\alpha = 0.45$)	703.8	−104.5	0.6614	−0.0574

Table 11: Each half of the domain scaled alone makes the fit *worse*; scaled together they help by $45\times$ the sum of the parts. Peaks are set by path-integrated conveyance along flow paths that cross both groups, so only domain-coherent changes drain them.

30 m.

The observable constrains one mode, not fifteen. A natural objection is that spatial redistribution should beat uniform scaling — that fifteen classes have freedom a single α does not. We tested it directly by scaling only part of the domain to $\alpha = 0.45$ and leaving the rest at the NLCD prior (Table 11).

Each half alone makes the fit *worse*, and the two together help by $45\times$ the sum of the parts — a strong positive interaction, not two competing signs. The mechanism is consistent with every other measurement: a mark’s peak is set by the *path-integrated* conveyance from upstream to that mark, and those paths cross both groups. Speeding part of a path does not drain it; the resulting fast/slow junction ponds water, so partial changes *raise* peaks. Only a domain-coherent change lowers them.

The conclusion is a design finding, not a tuning observation. The peak-WSE observable at these marks constrains *one* roughness degree of freedom at the prior width used here — the domain-scale, path-integrated conveyance — worth $\sim 10\%$ of the model–survey discrepancy, and about three at a $\pm 30\%$ width (Section 6.4). It does not carry fifteen classes’ worth of information. This unifies the earlier evidence rather than sitting beside it: the perfect-data twin recovers only the classes

Manning field	MAE	Δ	physically defensible?
NLCD lookup (uncalibrated)	0.7188	—	yes, by construction
uniform $\alpha = 0.70$	0.6894	-0.029	yes
uniform $\alpha = 0.30$	0.6392	-0.080	no (n to 0.008)
calibrated, 15 classes, $\sigma_n = 0.015$	0.6116	-0.107	mostly; one class at a bound
calibrated, 15 classes, $\sigma_\alpha = 0.30$	0.6154	-0.103	no; two classes at the floor
calibrated, 3 classes, $\sigma_n = 0.015$	0.6274	-0.091	no; two classes at the floor
calibrated, 3 classes, $\sigma_\alpha = 0.30$	0.6290	-0.090	no; two at the floor, one at $2.3\times$
calibrated, 3 classes, $\sigma_\alpha = 0.50$	0.6295	-0.089	no; all three at a bound

Table 12: Peak-WSE MAE against the 46 cluster-A marks. Both fifteen-class calibrations beat the entire uniform-roughness knob — including its physically indefensible end — and beat the best *defensible* uniform field by 0.078 m; spatial redistribution is worth about three and a half times what a single global scale can reach. The $\sigma_n = 0.015$ field took nine quasi-Newton iterations, the $\sigma_\alpha = 0.30$ field three, and they agree in misfit to four figures ($J_{\text{mis}} = 615.4$), which suggests a floor that roughness does not set. The three-class rows are the calibration Section 6.4 selects: their MAE moves by 0.002 m across a threefold change in prior width, and their two best-informed classes land on the floor under all three.

that move the domain mean, and the noisy twin pins the others to bounds precisely because they carry no signal to fit, so the optimizer fits noise in those directions instead.

Relation to equifinality. None of this is a new phenomenon: that many different parameter sets reproduce a hydrological observation about equally well is Beven’s equifinality thesis [22, 23], and the usual response is to characterize the acceptable set by sampling it. What an adjoint changes is the cost of the diagnosis. Where GLUE-style methods *sample* the equifinal set with many forward runs, the measurements above *measure* its dominant direction directly: four forward evaluations bound the authority of the whole roughness field, and two more show that the modes are not separable. The result is the same kind of statement, obtained at a cost that does not grow with the number of parameters — which is what makes it affordable before a calibration rather than after one.

Why this generalizes. The measurement uses nothing specific to roughness. Given a parameter, an observable, and a model error to explain, the same three steps — scan the dominant parameter direction, check whether sub-domain changes are separable, and compare the resulting authority against the residual — bound what calibrating that parameter can ever accomplish, before any optimizer is written. We suggest it as a routine precondition for calibration studies, which commonly report the misfit reduction achieved without establishing the ceiling it was approaching.

6.3 The calibration against the real survey

The scan sets the bar a multi-parameter calibration must clear to justify its parameters. Running the class calibration of Section 2.3 on the same configuration — fifteen NLCD classes, relative variables, prior width as discussed below, starting from the NLCD lookup — clears it (Table 12).

Two things qualify this number, one supporting and one limiting.

Supporting: the comparison that matters is against $\alpha = 0.70$, not $\alpha = 0.30$. The latter is the misfit’s unconstrained minimum and reaches $n = 0.008$, below any entry in the lookup; the former is the optimum implied by the same prior the calibration itself uses, and is the one-parameter

competitor that prior allows. Against it the calibration wins by 0.078 m, so the fifteen parameters are not decoration — *letting roughness vary by class removes error that no global scaling can*, which is a stronger statement than the scan alone justifies and the one the half-domain experiment (Table 11) would not have predicted.

Limiting, and there are three. First, the objective and the MAE do not move together. The calibrated field’s misfit is $J = 615.4$; sliding along the uniform scan’s measured MAE-versus- J relation to that same objective gives ≈ 0.596 m, where the calibrated field scores 0.6116, so redistribution buys objective slightly less MAE-efficiently than scaling does and the two should not be quoted interchangeably. That comparison is a counterfactual, though, and in the calibration’s favour: $J = 615.4$ lies below 673.7, the lowest objective the uniform knob reaches before the implicit solve fails, so the uniform field being compared against cannot actually be run. *The calibration reaches an objective global scaling cannot attain at all*. Second, the run is nine BLMVM iterations from the NLCD prior and was stopped by the 12-hour wall rather than by a tolerance: J fell 18.7% overall but only 0.01% on the last iteration while the gradient norm fell $8.5\times$, which is convergence behaviour without being a converged run, and developed-medium sits at its lower bound throughout; Section 2.3 discusses why that bound is partly an artifact of how the prior was stated. Third and most important, **0.6116 m is an in-sample score**: the 46 marks that define the objective are the 46 marks it is evaluated on. We report it as a fit, not as skill. Out-of-sample validation has to be cross-validation *within* the upstream band, because the middle band carries the drainage defect (Table 9) and would not test generalization so much as re-measure that defect.

Section 6.4 makes the second and third caveats concrete rather than conventional: the calibrated field lies 9.2 prior standard deviations from the NLCD lookup, and only 2.9% of that displacement is in a direction these marks constrain.

Set against the error budget: the model–survey discrepancy on these marks is 0.72 m and roughness, calibrated as well as we can currently calibrate it, accounts for 15% of it. The remaining 85% is rainfall forcing, mesh elevation, datum, and representation error at 30 m — and Section 7 shows that one of those is measurable with the same method, and larger.

6.4 How many parameters does the observable support?

Everything above infers the answer from scans. The eigenvalues of the prior-preconditioned Hessian give it directly. In coordinates where the prior is white, the posterior precision is $I + \Gamma^{1/2} H_{\text{mis}} \Gamma^{1/2}$, so each eigenvalue λ_i of the whitened misfit Hessian compares data information against prior information in one direction: the count with $\lambda_i > 1$ is the number of parameters the observable supports, and the eigenvectors say which *combinations* of classes those are — which no scan can.

Assembling it. With fifteen classes this needs no low-rank machinery, but it does need the right matrix. Differencing the *gradient* field fails here, and instructively: over a window of this length the objective is dense with wet/dry branch surfaces (Section 6.1), and while the adjoint is exact on its own branch the gradient jumps between branches, so a difference quotient samples a discontinuity. Our first attempt returned a matrix with 45% relative asymmetry and eigenvalues to -6.8 , which its own consistency check rejected. We therefore assemble the *Gauss–Newton* Hessian from observation sensitivities,

$$H = \frac{1}{\sigma^2} S^\top W S, \quad S_{ik} = \partial(\max_k H u)_i / \partial \alpha_k, \quad (7)$$

which differences the mark peaks rather than the gradient. It costs 16 forwards rather than 16 forward-plus-adjoints, and it is positive semi-definite by construction, so the failure mode that invalidated the first attempt cannot recur silently.

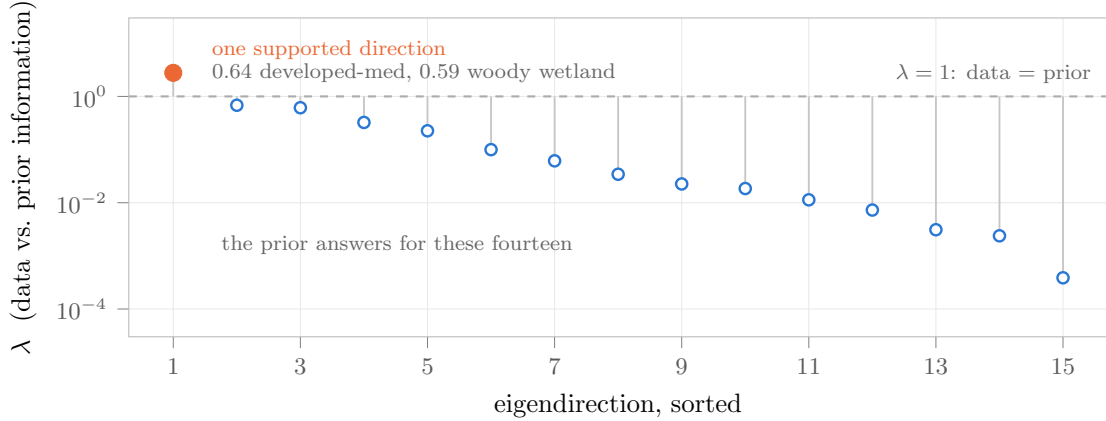


Figure 3: Eigenvalues of the prior-preconditioned Gauss–Newton Hessian over the fifteen land-cover classes, sorted. Above $\lambda = 1$ the marks constrain a direction more than the prior does; below it the prior sets the answer. One direction clears the line, and the remaining fourteen fall away over three decades — so a fifteen-parameter calibration is fitting one parameter and inheriting fourteen from its prior. The log axis is not cosmetic: on a linear one the cliff, which is the result, would be invisible.

Its own assumption — that a mark’s peak *time* is locally constant, so that the peak *value* is smooth in α — needs the same scrutiny, and on the production window it is not satisfied cleanly: 156 of 690 mark–column pairs (22.6%) move their peak time, against 27 (3.9%) on a two-event-hour pilot of the same configuration. The longer window is the reason. It admits more of each hydrograph, so a crest that was a monotone rising limb over two hours becomes a genuine, and therefore flat, maximum over twelve.

That flatness is what makes the movement harmless, and we checked rather than assumed it. Of the 156, 107 (71%) shift by a single 300-step observation sample, and the resulting change in the peak *value* has median 2.3 mm against a typical peak of 1.5 m — two parts in a thousand, the second-order $O(\Delta t^2 h'')$ cost of stepping one sample off a flat crest. Only ten pairs (1.6%) relocate by more than an hour, which is a genuinely different crest on a bimodal hydrograph. The distinction from the gradient-differencing attempt above is exact rather than rhetorical: there the differenced quantity *jumps* across a branch, whereas $\max_t h$ is *continuous* in α even where its argmax moves, since the two competing maxima are equal at the crossover. A secant of a continuous, piecewise-smooth function remains a valid finite difference; a secant across a discontinuity does not.

The answer is one. $\lambda = 2.78, 0.68, 0.61, 0.32, 0.23, 0.10, \dots$ (Figure 3) — a single eigenvalue above unity, a spectral gap of 4.1, and 2.20 degrees of freedom for signal in Rodgers’ sense. **These 46 marks support one roughness parameter** — one combination of the fifteen class values on which the data out-weighs the prior, and no second.

A shorter pilot of the same configuration (7,200 steps, two event hours) gives $\lambda = 3.03, 0.64, \dots$: the same count, the same three leading classes, and a slightly *larger* leading eigenvalue. We had expected the opposite, since a longer window lets each peak integrate more trajectory, and the reason it does not is worth stating. Over two event hours many marks have not crested, so the pilot’s “peak” is often a point on a rising limb, and a rising limb responds to roughness more sharply than a crest does — at a true maximum the first-order response in time has vanished by definition. The longer window measures the observable the calibration actually uses, and it is marginally less

informative, not more. Any argument that a longer experiment buys identifiability has to come from more marks crossing into the window, not from more trajectory per mark.

The scans and the spectrum agree, and they are independent: no optimizer, no line search, and no calibration enters Eq. (7).

But the scan varied nearly the wrong direction. The leading mode, with components -0.64 on developed-medium, $+0.59$ on woody wetland, -0.36 on developed-low and -0.23 on pasture, is dominated by the large-area classes. Its overlap with the uniform direction the α -scan varies — equal weight per *class* — is 0.218 , which for fifteen classes is indistinguishable from random (0.258). Its overlap with the same physical change weighted per *cell* is 0.699 . So the informative mode is the domain-scale conveyance mode we claimed, but area-weighted; the α -scan probes it only obliquely. That is why the class calibration beats uniform scaling by 0.078 m (Table 12): it is not that fifteen parameters beat one, it is that the scan’s one parameter was the wrong one.

What the calibration actually did. Projecting the calibrated displacement onto the eigenbasis is worth reporting, and it cuts both ways. The calibrated field sits **9.2 prior standard deviations** from the NLCD lookup, and of that motion only 2.9% lies in the single data-constrained direction; 85.6% lies in directions where data and prior are comparable and 11.5% where the prior alone determines the answer. Per class (Table 13) the marks shrink the prior uncertainty by 30% on developed-medium and 28% on woody wetland, by $11\text{--}20\%$ on three more, and by 5% or less on the remaining ten.

The reassuring half is the null space. Two iterations in, the optimizer had moved deciduous forest to $\alpha = 0.739$, mixed forest to 1.096 and cropland to 1.274 — three classes these marks teach nothing about — and by iteration nine accumulated curvature had returned all three to within 10% of the prior (0.983 , 1.006 , 1.098). Those excursions were transients of an under-informed quasi-Newton approximation rather than features of the minimizer: the misfit gradient in uninformative directions is not zero, merely uninformative, and given enough iterations the prior term dominates. A calibration stopped early would have reported them as results, which is an argument for running these problems to convergence even when the objective has visibly flattened.

The unreassuring half is what survives. The residual 9.2σ is concentrated in the two *best*-informed classes, and both are pressed against the edge of what the prior permits: developed-medium sits at its $\alpha = 0.3$ lower bound for all nine iterations, and woody wetland is driven to $\alpha = 0.355$, some 4.2σ from its lookup value. So the displacement is not motion in directions the data does not constrain. It lies in the one direction the data constrains, toward roughness below what the published table and our stated prior width jointly allow. Whether that value is admissible is not something the optimizer can settle. We put the lookup’s uncertainty at 30% or more, which admits it for the classes the marks constrain and, as the spectrum shows, releases twelve that they do not.

A direct test: calibrate three classes instead of fifteen. Everything above is a statement about a matrix. It makes a prediction an optimizer can falsify: if the marks really inform one direction, then calibrating only the classes that carry that direction should recover most of what all fifteen achieved — and if it does not, the information lives outside the leading eigenspace and the one-parameter reading is too narrow. We ran it. Taking the three classes with the largest components in the leading eigenvector (developed-medium, woody wetland, developed-low), freezing the other twelve at the prior, and calibrating from the NLCD lookup gives $J = 682.1$ after four iterations — the last two changing J by 0.02% and 0.001% . Against the NLCD prior’s 808.3 ,

NLCD	class	prior σ_n	posterior σ_n	learned
23	developed medium	0.015	0.0105	30%
90	woody wetland	0.015	0.0109	28%
81	pasture	0.015	0.0121	20%
22	developed low	0.015	0.0125	17%
21	developed open	0.015	0.0133	11%
95, 24, 11, 31		0.015	0.014–0.015	2–5%
41, 43, 82, 42, 71, 52		0.015	0.015	$\leq 1\%$

Table 13: What the 46 marks constrain in each land-cover class, as the fractional reduction from prior to posterior standard deviation, on the production-window spectrum. Ten classes narrow by 5% or less: any value a calibration reports for them is the prior, not the data.

three parameters remove 126.1 units where fifteen remove 150.9: **84% of the reduction from 20% of the parameters.** The twelve frozen classes were verified to sit bit-exactly at the prior in the output, so the comparison is what it claims to be.

The prediction survives the change of metric, which is the sharper test. Scored on peak-WSE MAE rather than on the objective it was fit to, the three-parameter field reaches 0.6274 m against the NLCD prior’s 0.7188 — 85% of the 0.107 m that fifteen parameters achieve, and within a hair of the 84% it recovered of the objective. The two agree here, and they did not for the fifteen-class field (Section 6.3): a fit confined to the directions the data constrains buys MAE at the same rate it buys objective, while one free to move all fifteen does not. Three of these parameters also beat the best physically defensible uniform field by 0.062 m, so nearly all of what class-by-class calibration achieves over global scaling is achieved by three classes.

Two details sharpen it further. The remaining 16% is real, not noise, and it decomposes exactly: the fifteen-parameter field pays 8.8 more units of prior penalty to buy 33.5 more units of misfit reduction. So the twelve non-leading classes are not literally decoration — in aggregate they reach a few per cent of J — but they are not where the information is. And the three parameters land where the fifteen-parameter calibration put them: developed-low at $\alpha = 1.626$ against 1.627, developed-medium on its $\alpha = 0.3$ bound in both. Two classes reproduced to three digits from a different starting point and a different active set is the behaviour of parameters the data actually determines. Woody wetland is the exception and is instructive: alone it dives to $\alpha = 0.301$ where the full calibration left it at 0.355, because with its partners frozen it must supply their share of a change the marks constrain only in combination — equifinality, exactly where the eigenvector says to expect it.

The same three classes under the fractional prior. The three-class run above used the absolute prior. Repeating it under the fractional prior at $\sigma_\alpha = 0.30$, the width we assign to the lookup, and at 0.50 as a bracket above it, gives 0.6290 and 0.6295 m (Table 12): 84% and 83% of the fifteen-class error reduction, and 90% and 94% of its objective reduction. The optimizer’s path is short. At $\pm 30\%$ the first step puts all three classes on a bound, exactly as it put seven of fifteen there (Section 2.3), and two more iterations bring developed-low back from $\alpha = 3$ to 2.34; at $\pm 50\%$ the first step lands on the corner of the box, the projected gradient there is zero, and the run converges in one iteration. The Gaussian prior is not what stops it: it contributes 15 units to J at $\pm 30\%$ and 10 at $\pm 50\%$ against a misfit near 630, so the bounds $\alpha \in [0.3, 3]$ do the constraining.

Three things follow. Developed-medium reaches $\alpha = 0.3$ under every prior and every active set we have run, and woody wetland does whenever its two partners are frozen (0.355 and 0.64 when they are not — the equifinality above), so in the three-class problem those values are set by

Marks	Supported	That data set is
46	1	the cluster-A marks (this study)
71	2	all admissible marks, 24-hour window
108	3	every QC-passed mark, 72-hour window
324	5	every Harvey mark in-domain, <i>before</i> QC
7×10^5	15	—

Table 14: How the supported-parameter count scales with survey size at fixed quality ($\sigma = 0.15$ m); a parameter is *supported* when the data narrows it more than the prior does ($\lambda > 1$). At fixed size, improving quality to $\sigma = 0.10$ m buys two more parameters (three in all) and $\sigma = 0.05$ m — better than any high-water-mark survey achieves — buys five. The table is a benchmark, not a forecast: it assumes independent, equal-variance residual errors and holds the measured per-mark sensitivity, the prior, and the linearization fixed, so the longer-window rows describe mark populations only, and a survey with correlated errors would enter through $\Gamma^{1/2} S^\top C^{-1} S \Gamma^{1/2}$ rather than through a count. That is the conservative reading and, as it turns out, the correct one: we expected a longer window to raise each mark’s sensitivity and it does not (see text).

the data, and they are $n = 0.036$ and 0.029 , below the lookup’s 0.040 for developed open land. Developed-low’s value is set by the prior: it moves from $1.63\times$ to $2.34\times$ to $3\times$ its lookup as the width opens while the MAE moves 0.002 m, which is the flat direction the eigenvector predicted. The three-parameter field therefore lies outside the prior as well. What the spectrum selects is the calibration the marks can determine, and what they determine is a roughness below the lookup’s range at any width we assign to it, which is evidence that the calibrated friction is compensating for an error that lives elsewhere (Section 7).

Would more marks fix it? This is the natural objection — 46 observations for 15 parameters sounds like too few — and Eq. (7) answers it, because λ scales linearly in the number of comparable observations and as σ^{-2} in their quality (Table 14). The answer is that more marks help, slowly, and that the supply is exhausted long before the parameterization is justified. Going from the 46 marks of this window to *all* 71 the model can legitimately be scored against supports two parameters. Every QC-passed mark in the domain, over a window long enough to catch all of them, supports three. Reaching five would take every Harvey mark that falls in the domain including the 62% that failed QC because they are channel surveys a 30 m cell cannot represent. Fifteen would take of order 10^5 marks.

The conclusion is therefore not that Manning roughness is unidentifiable. It is that *surveyed peak water-surface elevations at survey accuracy carry about two degrees of freedom of information about distributed roughness in this basin*; that this can be computed from sixteen forward runs before any optimizer is written; and that the limitation is a property of the observing system rather than of the method. A study that reported only its misfit reduction would have concluded the opposite.

7 Where else the error lives: the initial condition

Roughness, calibrated as well as we can calibrate it, explains 15% of this model’s error against the marks. Something explains the other 85%, and the candidates — rainfall forcing, mesh elevation, datum, representation error, the state the window starts from — are not all equally measurable. This section points the ceiling measurement of Section 6.2 at the one candidate it can probe with

IC scale a	J	MAE	Δ MAE per 0.1
0.80	6.652×10^2	0.6434	—
0.90	7.284×10^2	0.6796	+0.0362
1.00 (checkpoint)	8.083×10^2	0.7188	+0.0392
1.10	9.071×10^2	0.7610	+0.0422
1.20	1.025×10^3	0.8174	+0.0564

Table 15: Peak-WSE misfit against antecedent water storage. The response is monotone over the whole range with no interior optimum, straight to about 1% between $a = 0.8$ and 1.1 and convex above it: adding water hurts at an accelerating rate. The model holds systematically too much water at hour 29. No mark is dry at any point.

a single scan direction: the initial condition. The initial state is also the input a variational initialization would control, so the measurement doubles as a feasibility probe for initial-state estimation with the same adjoint machinery, which the conclusions take up.

Our window begins at event hour 29 from a checkpoint of the 72-hour forward, and the natural single direction to scan is the amount of water that checkpoint holds. Scaling all three conserved variables by a common factor a changes antecedent storage while leaving every velocity unchanged, since $u = (hu)/h$ is invariant. Each point is one forward (Table 15).

Three things follow, and they matter more than the roughness result.

The initial state has about four times the authority of roughness (Figure 2). Per unit fractional change, d MAE is 0.392 for the initial condition over its well-sampled range ($a \in [0.8, 1.1]$) against 0.098 for a uniform roughness scale. Like for like at a 20% perturbation: 0.075 m versus 0.020 m. The plausibility runs the same way: a 20% error in water stored after a 29-hour spin-up under radar-derived rainfall is within the uncertainty of that forcing [21], while Manning values at 30% of the published table lie below the table’s own value for open land.

The response has the opposite curvature to roughness’s. $J(\alpha)$ for roughness is concave and saturating — the knob runs out — consistent with a parameter acting indirectly through path-integrated conveyance. $J(a)$ for the initial state is straight to about 1% between $a = 0.8$ and 1.1 and *convex* above, steepening as water is added, and it does not turn over anywhere in the range. Extending the scan below the table confirms it: at $a = 0.6$ — a 40% reduction in antecedent water — the MAE is still falling, to 0.5818 m (Figure 2). Neither the slope nor the sign gives any indication of an optimum near the checkpoint value. That is the signature of a systematic bias rather than a poorly-tuned parameter: the model does not hold slightly the wrong *distribution* of water at hour 29, it holds too much of it.

It is a diagnosis, not a knob. We are not proposing to calibrate the initial condition against these marks. Over this window the initial state carries 3 unknowns on each of 2.93M cells against 46 peak observations, and everything Section 6.2 establishes about identifiability applies with three more orders of magnitude of freedom: such a calibration would fit the marks exactly and mean nothing. What the scan does establish is the weaker and more useful claim that the residual these marks measure lives substantially in the water balance — in the rainfall forcing or the antecedent state — rather than in the friction, which is where a roughness calibration alone would have kept looking.

One caveat bounds all three. The checkpoint is a *model* state, not an observation, so it already contains accumulated forcing and elevation error; the scan perturbs that state rather than an independent measurement of it. A calibration of the initial condition would therefore absorb such error rather than correct it, and could quietly launder the downstream drainage defect of Section 6.1

into a “corrected” initial state — the same trap as scoring against the 37 marks the model cannot drain. Establishing whether a 20% water-balance error at hour 29 is physically plausible is a question for the forcing data, not for the optimizer. Our answer is that it is: the antecedent state after a 29-hour spin-up under radar rainfall is among the least certain inputs the model has, and an error of that size in it is plausible. The uncertainty study of this event in [21] finds precipitation and mesh resolution to dominate the simulated inundation over the choice of outlet boundary condition [21], which points the same way.

8 Conclusions and outlook

An exact, verified Jacobian turns out to be the single artifact from which everything else follows cheaply: adjoint state and parameter sensitivities, gradient-based calibration, and implicit stepping. The capability is production-ready in the sense that matters for an operational code — behind a flag, FD-gated in CI, and with the verification baseline (FD coloring) kept alive in the configuration system.

The second artifact is a measurement rather than a capability, and it is the one we would most like others to copy. Asking what fraction of a model’s error a parameter is *able* to explain is cheap, optimizer-independent, and it reframes what a calibration result means. Here it bounded roughness at 4% of the error within defensible values and 11% at the edge of plausibility; the calibration then reached 15%, beating every global scaling and confirming that letting roughness vary by class removes error a single global factor cannot. What makes the measurement worth copying is not that number but the two it led to. Pointed at the residual instead of the parameter, the same machinery isolated a reach the model cannot drain; pointed at the initial condition, it found more authority than roughness has. A calibration study that had only reported its misfit reduction would have found neither.

Next steps, in order. First, cross-validate the calibration reported here — nine quasi-Newton iterations stopped by a wall-clock limit, with one class on a bound, is a fit, not demonstrated skill — and the held-out marks must come from within the upstream band, because the middle band carries the drainage defect. Second, calibrate the number of parameters the observable supports rather than the number the land-cover map offers; the spectrum of the prior-preconditioned Hessian would settle that count directly, and at fifteen parameters it is a small dense eigenproblem rather than a research project. Third, the water balance: the initial-condition scan says the antecedent state carries more of this residual than the friction does, which makes the forcing and the initial condition — not the roughness — the next thing to calibrate or diagnose, with the machinery applying unchanged and dJ/du_0 already verified. Fourth, the conveyance and outlet representation the drainage finding implicates. Finally, coupling the gradient path to PyTorch through the existing PETSc–PyTorch bridge for learned-physics and hybrid data-assimilation experiments.

A closing note on cost. Checkpointing makes the long-window adjoint memory-feasible, but at the 30 m mesh’s CFL-limited $\Delta t = 0.25$ s a full 36-hour Hurricane Harvey gradient still costs several forward sweeps over half a million steps. The affordable path stacks reductions that are all now in place. First, fully implicit stepping at CFL-free time steps: backward Euler with the assembled Jacobian carries the 30 m, 2.93M-cell Harvey-region mesh at $\Delta t = 1$ s — a step the differentiable explicit path cannot take at any tested size (Section 4) — with Newton converging in about two iterations per step behind a block-Jacobi-preconditioned Krylov solve at converged inner tolerances; larger implicit steps are under investigation. Second, GPU execution of the entire gradient pipeline: the right-hand side, the assembled Jacobian and its transpose solves, the preconditioner setup, the parameter Jacobian, the observation operator, and the revolve-checkpointed trajectory

are all device-resident. On four A100 GPUs, a forward-plus-adjoint gradient of a one-simulated-hour window on that mesh (3,600 implicit steps, 400 in-memory checkpoints) takes about nine minutes, and one TAO iteration of a dense-gauge twin calibration (418,076 gauges, 39-step window) takes 1.7s — roughly $19\times$ one 64-core CPU node of the same system. Device and host runs produce bitwise-identical trajectories under a common compiler and gradients identical to every printed digit, so the finite-difference gates of Table 1 carry over unchanged. Third, land-cover-class parameterization (tens of TAO evaluations instead of hundreds) and calibration windows placed on the storm peak rather than the full event.

Code and data availability

RDycore is open source at <https://github.com/RDycore/RDycore>; the adjoint, Jacobian, implicit-stepping and calibration capability of this paper is on the `adams/gpu-implicit` branch behind the `numerics.jacobian` and `adjoint` configuration flags, and the verification gates of Table 1 run in its `ctest` suite. The exact revision, the campaign scripts, the observation tables, and the per-mark peak dumps behind Figures 2 and 3 will be archived on Zenodo at submission [**DOI to be minted**]. Hurricane Harvey high-water marks are the USGS Short-Term Network survey, retrieved through the STN Flood Event Viewer API (<https://stn.wim.usgs.gov>) with the retrieval script and the filtered mark table in the archive; stage records are USGS NWIS; rainfall is MRMS. [**Mesh, checkpoint, and rainfall archive: location to be stated.**]

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